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Home Assignment №1

Due on October 3, 2026, 23:59 PM

Exercise 1

[5 points]. This problem reviews basic concepts from probability.

a) [1 point]. A biased die has the following probabilities of landing on each face:

face	1	2	3	4	5	6
P(face)	.1	.1	.2	.2	.4	0

I win if the die shows even. What is the probability that I win? Is this better or worse than a fair die (i.e., a die with equal probabilities for each face)?

b) [1 point]. Recall that the expected value $\mathbb{E}[X]$ for a random variable X is

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} p(X = x)\ x,$$

where $\mathcal{X}$ is the set of values X may take on. Similarly, the expected value of any function f of random variable X is

$$\mathbb{E}[f(X)] = \sum_{x \in \mathcal{X}} p(X = x)\ f(x).$$

Now consider the function below, which we call the "indicator function"

$$\mathbb{I}[X = a] := \begin{cases} 1 & \text{if } X = a \\ 0 & \text{if } X \neq a \end{cases}.$$

Let X be a random variable which takes on the values $3, 8$ or 9 with probabilities p_3, p_8 and p_9 respectively. Calculate $\mathbb{E}[\mathbb{I}[X = 8]]$.

c) [2 points]. Recall the following definitions:

- Entropy: $H(X) = -\sum_{x\in\mathcal{X}} p(X=x)\log_2 p(X=x) \;\; = -\mathbb{E}[\log_2 p(X)]$
- Joint entropy: $H(X,Y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(X=x,Y=y)\log_2 p(X=x,Y=y) \;\; = -\mathbb{E}[\log_2 p(X,Y)]$
- Conditional entropy: $H(Y|X) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(X=x,Y=y)\log_2 p(Y=y|X=x) \;\; = -\mathbb{E}[\log_2 p(Y|X)]$
- Mutual information: $I(X;Y) = \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(X=x,Y=y)\log_2 \frac{p(X=x,Y=y)}{p(X=x)p(Y=y)}$

Using the definitions of the entropy, joint entropy, and conditional entropy, prove the following chain rule for the entropy:

$$H(X,Y) = H(Y) + H(X|Y).$$

d) [1 point]. Recall that two random variables X and Y are *independent* if

$$\text{for all } x\in\mathcal{X} \text{ and all } y\in\mathcal{Y}, \quad p(X=x,Y=y) \;=\; p(X=x)p(Y=y).$$

If variables X and Y are independent, is $I(X;Y)=0$? If yes, prove it. If no, give a counter example.

Exercise 2

[4 points]. Given a training set $\mathcal{D} = \{(x^{(i)}, y^{(i)}), i = 1,\ldots,M\}$, where $x^{(i)} \in \mathbb{R}^N$ and $y^{(i)} \in \{1,2,\ldots,C\}$, derive the maximum likelihood estimates of the naive Bayes for real valued $x_j^{(i)}$ modeled with a Laplacian distribution, *i.e.*,

$$p(x_j|y=c) = \frac{1}{2\sigma_{j|c}}\exp\left(-\frac{|x_j-\mu_{j|c}|}{\sigma_{j|c}}\right).$$

Exercise 3

[4 points]. Prove that in binary classification, the posterior of linear discriminant analysis, *i.e.*, $p(y=1|x;\varphi,\mu,\Sigma)$, is in the form of a sigmoid function

$$p(y=1|x;\theta) = \frac{1}{1+e^{-\theta^T x}}$$

where θ is a function of $\{\varphi,\mu,\Sigma\}$. Hint: remember to use the convention of letting $x_0 = 1$ that incorporates the bias term into the parameter vector θ.

Exercise 4

[2 points]. Let

$$P_1 = \mathcal{N}(x; \mu_1, \Sigma), \qquad P_2 = \mathcal{N}(x; \mu_2, \Sigma)$$

be two N-dimensional Gaussian distributions with the same positive-definite covariance matrix Σ. Starting from

$$D_{\mathrm{KL}}(P_1 \| P_2) = \mathbb{E}_{P_1}\left[\log \frac{P_1(x)}{P_2(x)}\right],$$

derive a closed-form expression for the divergence and show that, in this shared-covariance case,

$$D_{\mathrm{KL}}(P_1 \| P_2) = D_{\mathrm{KL}}(P_2 \| P_1).$$

You may use $\mathbb{E}_{P_1}[x] = \mu_1$ and $\mathbb{E}_{P_1}[(x-\mu_1)(x-\mu_1)^T] = \Sigma$.

Exercise 5

[2 points]. Let $A = A^T \in \mathbb{R}^{N \times N}$ and suppose $A \succeq 0$, meaning

$$x^T A x \geq 0 \qquad \text{for every } x \in \mathbb{R}^N.$$

Prove the following two properties of positive semidefinite matrices:

a) Show that $A_{ii} \geq 0$ and

$$A_{ij}^2 \leq A_{ii} A_{jj} \qquad \text{for all } i, j.$$

b) For every matrix $B \in \mathbb{R}^{N \times K}$, $B^T A B \succeq 0$.

1. ⓐ $X \in \{2,4,6\}$

$p(\text{win}) = p(x=2) + p(x=4) + p(x=6) = 0.1 + 0.2 + 0 = 0.3$

$= 0.3$

fair die : $p(x=n) = \frac{1}{6}$

$p(\text{win}) = p(x=2) + p(x=4) + p(x=6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$

$P_{\text{fair}}(\text{win}) = \frac{1}{2}$

$0.3 < 0.5$

∴ The biased die is worse for me

ⓑ $I[X=a] = \begin{cases} 1, & X=a \\ 0, & X \neq a \end{cases}$

$x \in \{3,8,9\}$

$E[I[X=8]]$

$E[f(x)] = \sum_x p(X=x) f(x)$

$E[I(X=8)] = p_3 I[3=8] + p_8 I[8=8] + p_9 I[9=8]$

$= 0p_3 + 1p_8 + 0p_9$

$E[I[X=a]] = p_8$

ⓒ $H(X,Y) = -\sum_x \sum_y p(x,y) \log_2 p(x,y)$ $\quad p(x,y) = p(x|y)p(y)$

$\log p(x,y) = \log[p(x|y)p(y)]$

$= \log p(x|y) + \log p(y)$

$H(X,Y) = -\sum_x \sum_y p(x,y)[\log p(x|y) + \log p(y)] = -\sum_x \sum_y p(x,y) \log p(x|y) - \sum_x \sum_y p(x,y) \log p(y)$

$H(Y) = -\sum_x \sum_y p(x,y) \log p(y)$ $\quad \because \sum_x p(x,y) = p(y)$ $\therefore -\sum_y p(y) \log p(y) = H(Y)$

$H(X|Y) = -\sum_x \sum_y p(x,y) \log p(x|y)$

$H(X,Y) = H(Y) + H(X|Y)$

ⓓ $\because I(X;Y) = 0$ Yes

$\because p(x,y) = p(x)p(y)$

$\frac{p(x,y)}{p(x)p(y)} = 1$

$I(X;Y) = \sum_x \sum_y p(x,y) \log 1 = 0$

$X \perp Y \Rightarrow I(X;Y) = 0$

2.

① $p(x|y=c) = \prod_{j=1}^{N} p(x_j|y=c)$ $\therefore p(x,y) = p(y)\,p(x|y) = p(y)\prod_{j=1}^{N} p(x_j|y)$

$\therefore p(y=c)$, $\mu_{j|c}$ $\sigma_{j|c}$

② $M_c = \sum_{i=1}^{M} I[y^{(i)}=c]$

$$L(\phi) = \prod_{i=1}^{M} \phi_{y^{(i)}}$$

$$\ell(\phi) = \sum_{i=1}^{M} \log \phi_{y^{(i)}}$$

$$\ell(\phi) = \sum_{c=1}^{C} M_c \log \phi_c \qquad \sum_c \phi_c = 1$$

$$\mathcal{L} = \sum_c M_c \log \phi_c + \lambda\left(\sum_c \phi_c - 1\right)$$

$$\frac{\partial \mathcal{L}}{\partial \phi_c} = \frac{M_c}{\phi_c} + \lambda = 0$$

$$\phi_c = -\frac{M_c}{\lambda} \to -\frac{1}{\lambda}\sum_c M_c = 1 \qquad \because \sum_c M_c = M$$

$$\lambda = -M$$

$$\therefore \hat{\phi}_c = \frac{M_c}{M}$$

③

$$L(\mu_{j|c}, \sigma_{j|c}) = \prod_{i:y^{(i)}=c} \frac{1}{2\sigma_{j|c}} \exp\left[-\frac{|x_j^{(i)} - \mu_{j|c}|}{\sigma_{j|c}}\right]$$

$$\ell = -M_c \log(2\sigma_{j|c}) - \frac{1}{\sigma_{j|c}} \sum_{i:y^{(i)}=c} |x_j^{(i)} - \mu_{j|c}|$$

$$\sum_{i:y^{(i)}=c} |x_j^{(i)} - \mu_{j|c}|$$

$$\hat{\mu}_{j|c} = \text{median}\{x_j^{(i)} : y^{(i)} = c\}$$

④

$$\ell = -M_c \log(2\sigma) - \frac{1}{\sigma}\sum_i |x_i - \mu| \qquad S = \sum_{i:y^{(i)}=c} |x_j^{(i)} - \mu_{j|c}|$$

$$\ell = -M_c \log(2\sigma) - \frac{S}{\sigma} \qquad \frac{\partial \ell}{\partial \sigma} = -\frac{M_c}{\sigma} + \frac{S}{\sigma^2} \qquad -\frac{M_c}{\sigma} + \frac{S}{\sigma^2} = 0$$

$$-M_c\sigma + S = 0 \qquad \sigma = \frac{S}{M_c}$$

$$\hat{\sigma}_{j|c} = \frac{1}{M_c}\sum_{i:y^{(i)}=c} |x_j^{(i)} - \hat{\mu}_{j|c}|$$

$$\therefore \hat{\phi}_c = \frac{M_c}{M} \qquad \hat{\mu}_{j|c} = \text{median}\{x_j^{(i)} : y^{(i)} = c\}$$

$$\hat{\sigma}_{j|c} = \frac{1}{M_c}\sum_{i:y^{(i)}=c} |x_j^{(i)} - \hat{\mu}_{j|c}|$$

3. $P(y=1|x;\theta) = \frac{1}{1+e^{-\theta^T x}}$

①

LDA: $x|y=1 \sim N(\mu_1, \Sigma)$ $\quad P(y=1) = \phi$

$x|y=0 \sim N(\mu_0, \Sigma)$ $\quad P(y=0) = 1-\phi$

② $P(y=1|x) = \frac{P(x|y=1)P(y=1)}{P(x|y=1)P(y=1) + P(x|y=0)P(y=0)}$

odds: $\frac{P(y=1|x)}{P(y=0|x)} = \frac{P(x|y=1)\phi}{P(x|y=0)(1-\phi)}$

$\log \frac{P(y=1|x)}{P(y=0|x)} = \log \frac{\phi}{1-\phi} + \log \frac{P(x|y=1)}{P(x|y=0)}$

③ $P(x|y=c) = \frac{1}{(2\pi)^{\frac{N}{2}}|\Sigma|^{\frac{1}{2}}} \exp\left[-\frac{1}{2}(x-\mu_c)^T \Sigma^{-1}(x-\mu_c)\right]$

$\log \frac{P(x|y=1)}{P(x|y=0)} = -\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_0)^T \Sigma^{-1}(x-\mu_0)$

$(X-\mu_1)^T \Sigma^{-1}(x-\mu_1) = x^T\Sigma^{-1}x - 2\mu_1^T\Sigma^{-1}x + \mu_1^T\Sigma^{-1}\mu_1$

$x^T\Sigma^{-1}x - 2\mu_0^T\Sigma^{-1}x + \mu_0^T\Sigma^{-1}\mu_0$

$\therefore \log \frac{P(y=1|x)}{P(y=0|x)} = \log \frac{\phi}{1-\phi} + (\mu_1-\mu_0)^T\Sigma^{-1}x$

$= -\frac{1}{2}(\mu_1^T\Sigma^{-1}\mu_1 - \mu_0^T\Sigma^{-1}\mu_0)$

④ $\theta_x = \Sigma^{-1}(\mu_1-\mu_0)$ $\quad \theta_0 = \log \frac{\phi}{1-\phi} - \frac{1}{2}(\mu_1^T\Sigma^{-1}\mu_1 - \mu_0^T\Sigma^{-1}\mu_0)$

$\log \frac{P(y=1|x)}{P(y=0|x)} = \theta_0 + \theta_x^T x$ $\quad P(y=0|x) = 1 - P(y=1|x)$

$\log \frac{p}{1-p} = \theta_0 + \theta_x^T x$ $\quad p = \frac{1}{1+e^{-(\theta_0+\theta_x^T x)}}$

when $x_0 = 1$

$\theta = \begin{bmatrix} \theta_0 \\ \theta_x \end{bmatrix}$ $\quad x = \begin{bmatrix} 1 \\ x \end{bmatrix}$

$P(y=1|x) = \frac{1}{1+e^{-\theta^T x}}$

4.

① $P_1(x) = \frac{1}{(2\pi)^{\frac{N}{2}}|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)}$

$P_2(x) = \frac{1}{(2\pi)^{\frac{N}{2}}|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)}$

$\log\frac{P_1(x)}{P_2(x)} = -\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)$

$D_{KL}(P_1\|P_2) = \frac{1}{2}E_{P_1}[(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)] - \frac{1}{2}E_{P_1}[(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)]$

② $E_{P_1}[(x-\mu_1)(x-\mu_1)^T] = \Sigma$

$E_{P_1}[(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)] = tr[\Sigma^{-1}E[(x-\mu_1)(x-\mu_1)^T]]$

$\therefore \frac{N}{2} = tr(\Sigma^{-1}\Sigma) = tr(I) = N$

③ $x-\mu_2 = (x-\mu_1) + (\mu_1-\mu_2)$ $\quad \delta = \mu_1-\mu_2$

$x-\mu_2 = (x-\mu_1) + \delta$

$(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)$

$= (x-\mu_1)^T\Sigma^{-1}(x-\mu_1) + 2\delta^T\Sigma^{-1}(x-\mu_1) + \delta^T\Sigma^{-1}\delta$

$\because E_{P_1}[x-\mu_1] = 0$

$\therefore E_{P_1}[(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)] = N + \delta^T\Sigma^{-1}\delta$

$D_{KL}(P_1\|P_2) = \frac{1}{2}[N + \delta^T\Sigma^{-1}\delta] - \frac{N}{2}$

$\therefore D_{KL}(P_1\|P_2) = \frac{1}{2}(\mu_1-\mu_2)^T\Sigma^{-1}(\mu_1-\mu_2)$

④ $D_{KL}(P_2\|P_1) = \frac{1}{2}(\mu_2-\mu_1)^T\Sigma^{-1}(\mu_2-\mu_1)$

$\mu_2-\mu_1 = -(\mu_1+\mu_2)$

$(-a)^T\Sigma^{-1}(-a) = a^T\Sigma^{-1}a$

$\therefore D_{KL}(P_2\|P_1) = \frac{1}{2}(\mu_1-\mu_2)^T\Sigma^{-1}(\mu_1-\mu_2)$

$D_{KL}(P_1\|P_2) = D_{KL}(P_2\|P_1)$

5.

ⓐ $x = e_i$.

$\because A \geq 0$

$\therefore e_i^T A e_i \geq 0$ $\because e_i^T A e_i = A_{ii}$ $\therefore A_{ii} \geq 0$

$x = e_i + te_j$ $\because x \geq 0$

$\therefore x^T A x \geq 0$

$(e_i + te_j)^T A (e_i + te_j) \geq 0$

$A_{ii} + 2tA_{ij} + t^2 A_{jj} \geq 0 \rightarrow \Delta \leq 0$

$(2A_{ij})^2 - 4A_{jj}A_{ii} \leq 0$

$4A_{jj}A_{ii} \geq 4A_{ij}$

$A_{ij}^2 \leq A_{ii}A_{jj}$

ⓑ $B \in R^{N \times k}$

$\because A = A^T$ $\therefore (B^T A B)^T = B^T A^T B$

$\because A^T = A$ $\therefore (B^T A B)^T = B^T A B$

$\therefore$ Symmetric.

$x^T B^T A B x = (Bx)^T A (Bx)$ $z = Bx$

$\because A \geq 0$ $\therefore z \in R^N$ $z^T A z \geq 0$

$\therefore (Bx)^T A (Bx) \geq 0$

$x^T B^T A B x \geq 0$

$\therefore B^T A B \geq 0$